

DATA WAREHOUSING & DATA MINING

Data Warehouse Architecture

Understanding the layered structure that turns raw data into decision-ready information

What Is a Data Warehouse?

A subject-oriented, integrated, time-variant, and non-volatile collection of data used to support management decision-making.

1

Subject-Oriented

Organized around key business subjects like sales, customers, or products rather than day-to-day operations.

2

Integrated

Consolidates data from multiple, often inconsistent, source systems into one unified format.

3

Time-Variant

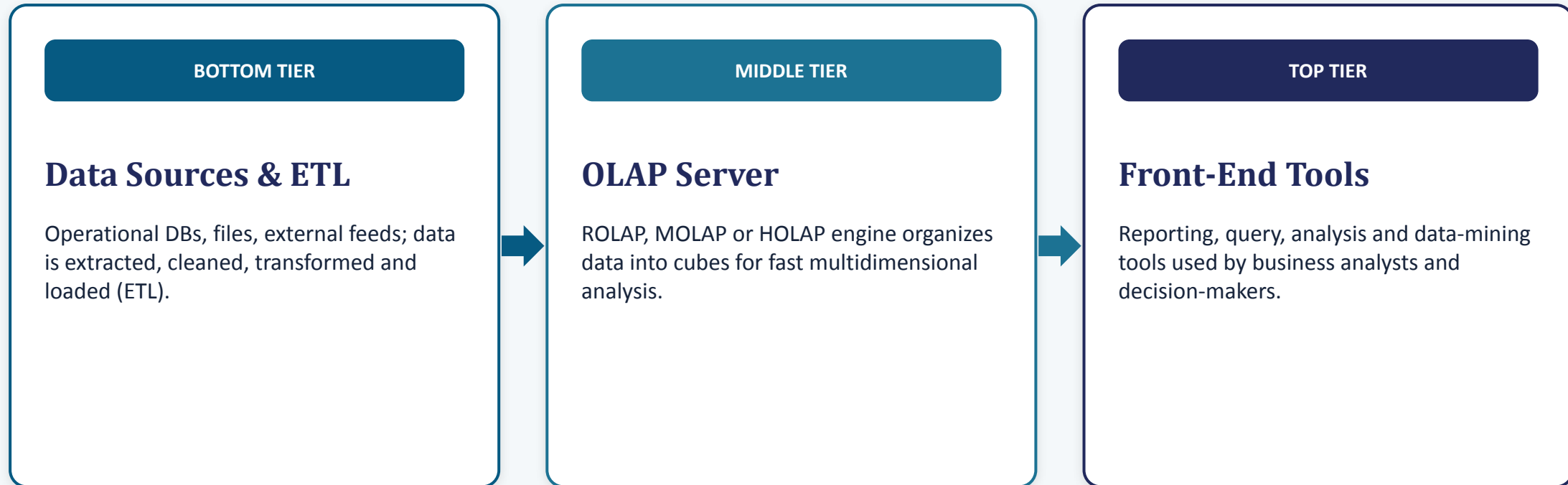
Stores historical data over long periods, enabling trend analysis across time.

4

Non-Volatile

Once loaded, data is stable and read-only — it is not overwritten or deleted with each transaction.

Three-Tier Data Warehouse Architecture



Data flows bottom → top: from raw sources, through consolidation and modeling, to the tools decision-makers use every day.

Key Components of the Architecture

1

ETL Process

Extract, Transform, Load — pulls data from heterogeneous sources, cleans and reshapes it, then loads it into the warehouse.

2

Metadata Repository

Stores data about the data — source, structure, transformation rules — enabling traceability and governance.

3

OLAP Engine

Supports multidimensional views of data via cubes, enabling slicing, dicing, drill-down and roll-up operations.

4

Data Marts

Smaller, department-focused subsets of the warehouse tailored to specific business units like sales or finance.

Why the Architecture Matters

A well-layered warehouse turns scattered operational data into a single source of analytical truth.

Faster Decisions

Pre-aggregated, modeled data lets analysts query in seconds, not hours.

Consistent View

One integrated source removes conflicting numbers across departments.

Historical Insight

Time-variant storage reveals trends that single transactional systems miss.

Foundation for Mining

Clean, structured warehouse data is the raw material data-mining algorithms need.